

Project Harbour

Retirement drawdown risk and what it means for allocation

Monte Carlo simulation. 20,000 paths per scenario.

A normal returns test is too kind

15.2%

Textbook case. Normal returns and no health bills.

24.9%

Clustered bad years, health shocks and tax.

40.0%

Same, plus a bear market in year one.

Chance the money runs out before age 95.

Same retiree each time. \$40,000 a year after tax, from a \$1,000,000 pot, at 60/40. Only the realism changes.

What was simulated

Markets

Three regimes. Expansion, bear and stagflation. Bad years cluster, so a retiree cannot wait them out.

Health

Long term care from age 75. About 45% of paths face it, at about \$193,000. Smaller one off bills too.

Spending rules

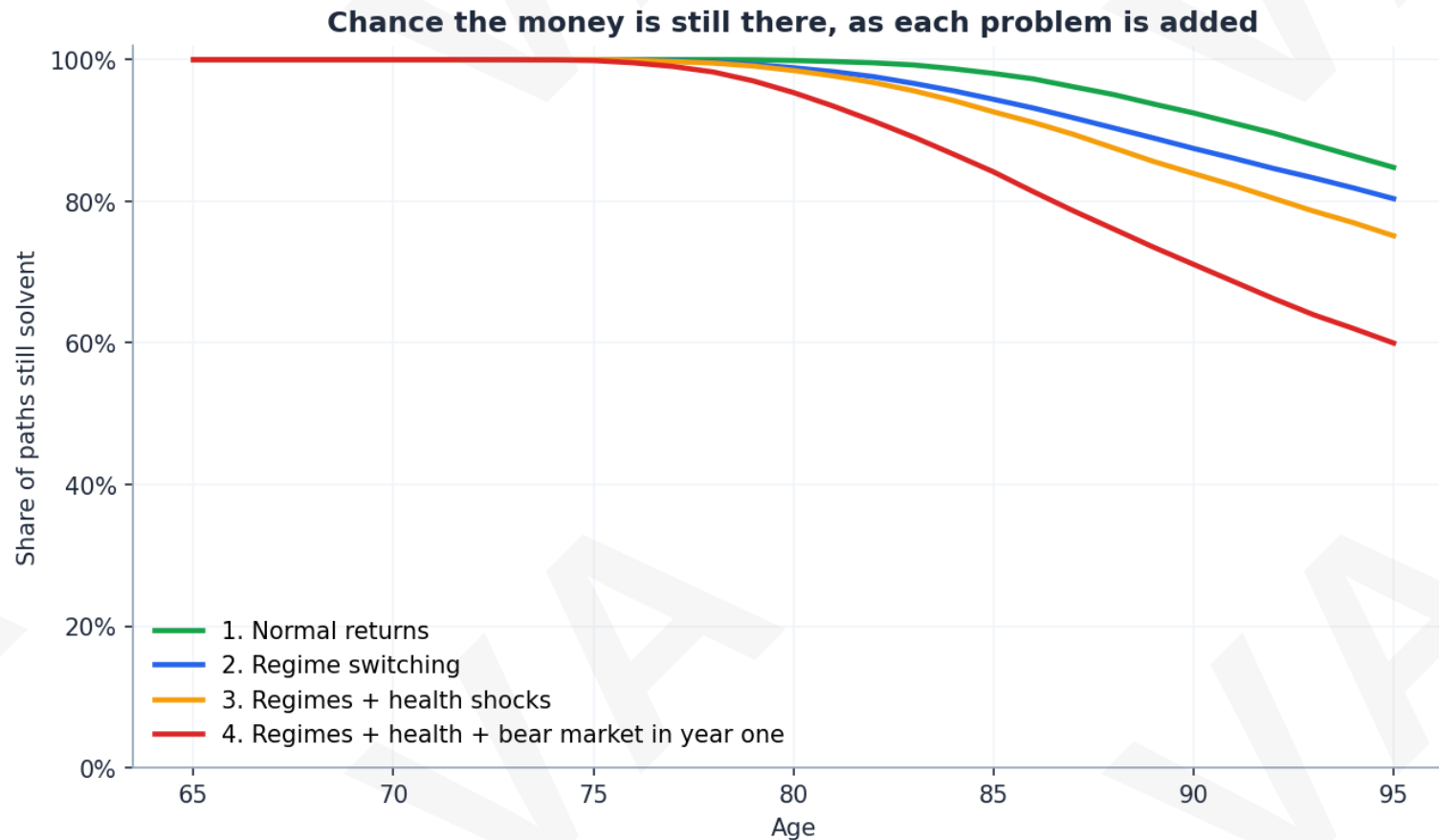
Static. Guardrails that cut or raise spending by 10%. Variable percentage withdrawal.

Tax

Taxable, traditional and Roth accounts. Four withdrawal orders. Forced withdrawals from age 73.

\$1,000,000 pot. Age 65 to 95. \$40,000 a year after tax. 60/40 base mix. All figures in today's money.

Each added problem pushes the ruin rate up



Regimes add 4 points

Same average return. Bad years now arrive together.

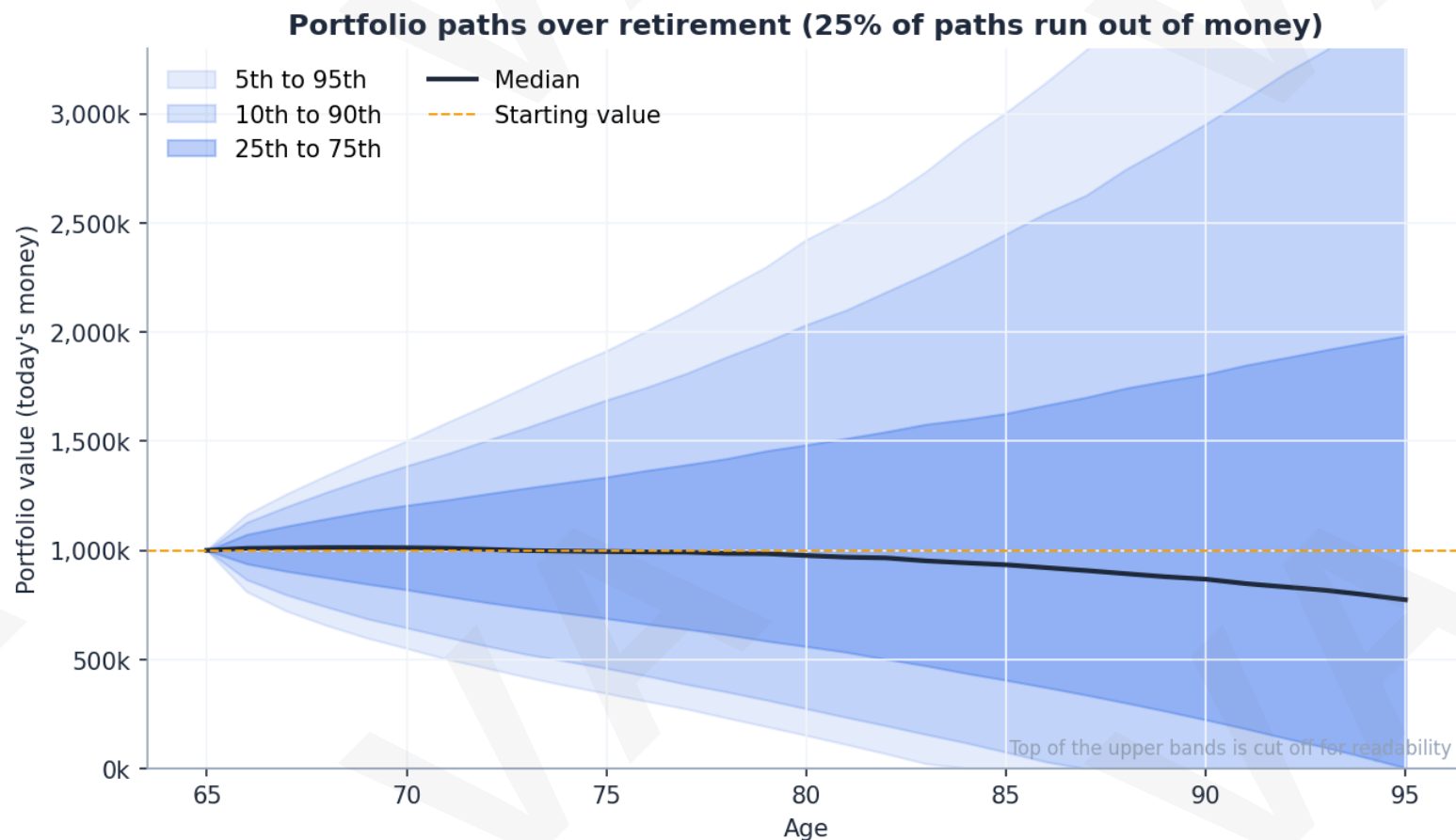
Health adds 5 points

A care bill lands when the pot is already thin.

A year one bear adds 15 points

One bad year at the start does the most damage.

The median path survives. One in four does not



\$773,259

Median ending wealth. Close to where it began.

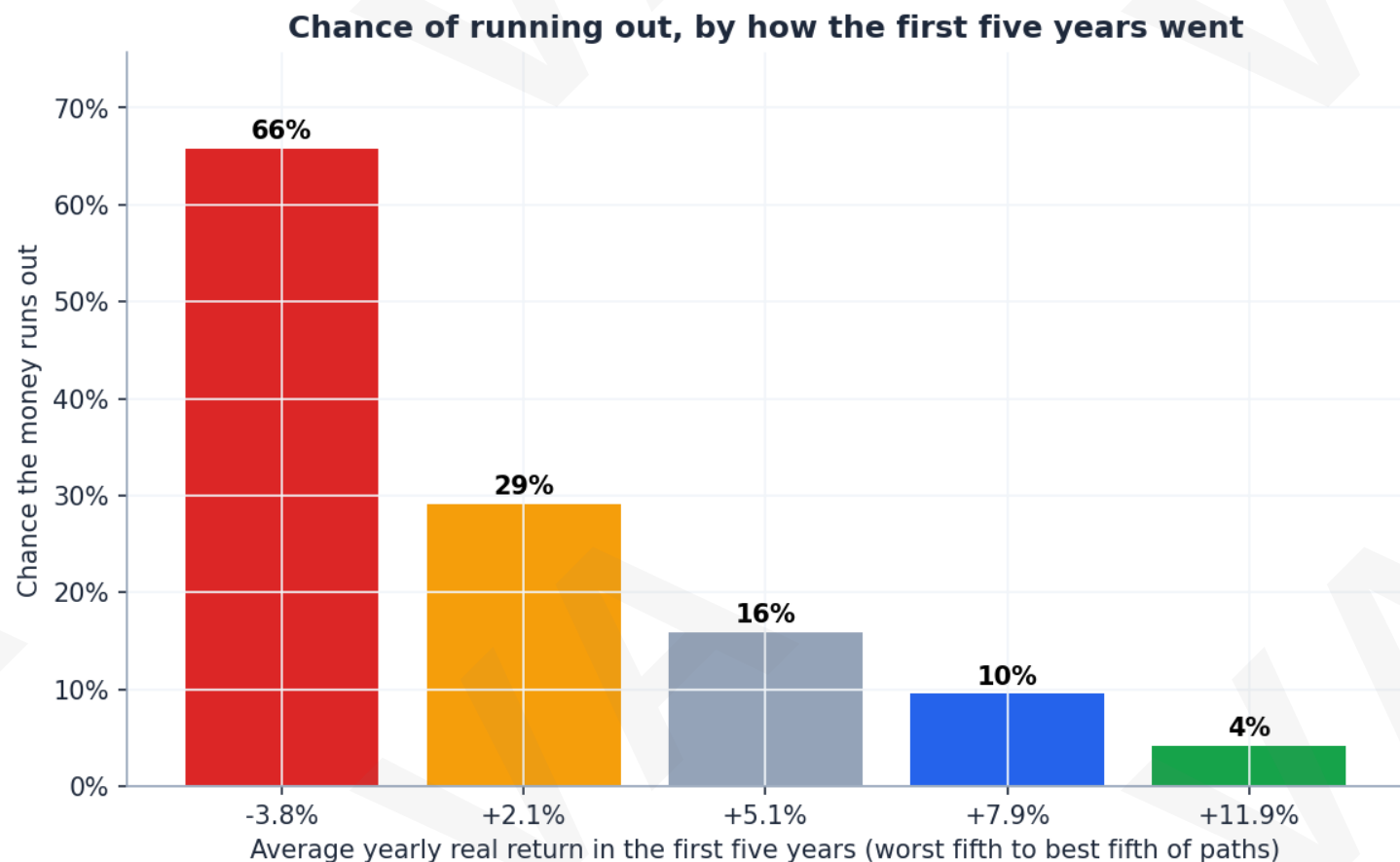
\$1,360,412

Expected ending wealth. A few paths do very well, so the average misleads.

55.9%

Chance of ending below the starting value.

The first five years matter more than the stock weight



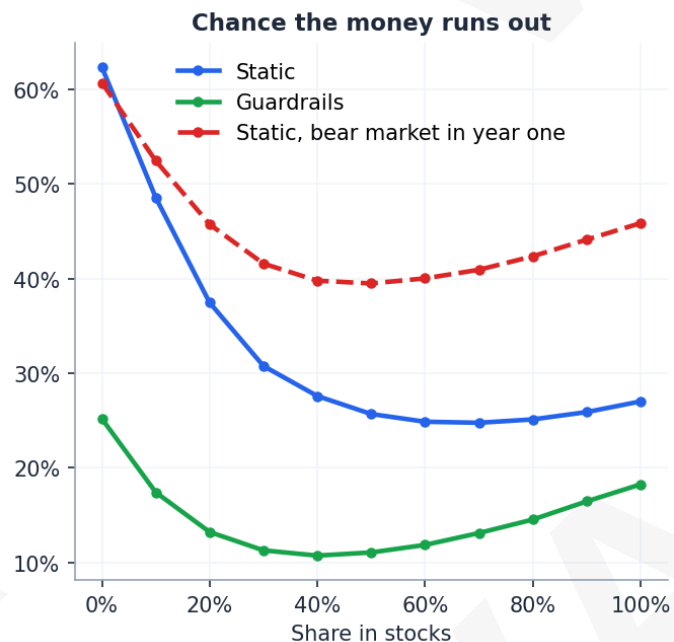
Worst fifth. 66% run out

These paths averaged -3.8% a year for five years.

Best fifth. 4% run out

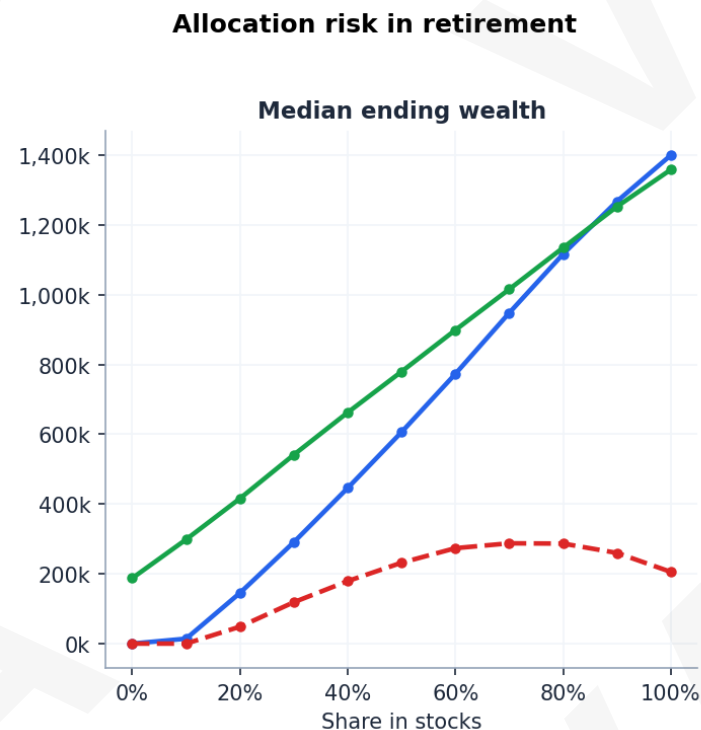
These paths averaged +11.9% a year. Same withdrawals.

Ruin risk is U shaped. The middle is wide and flat



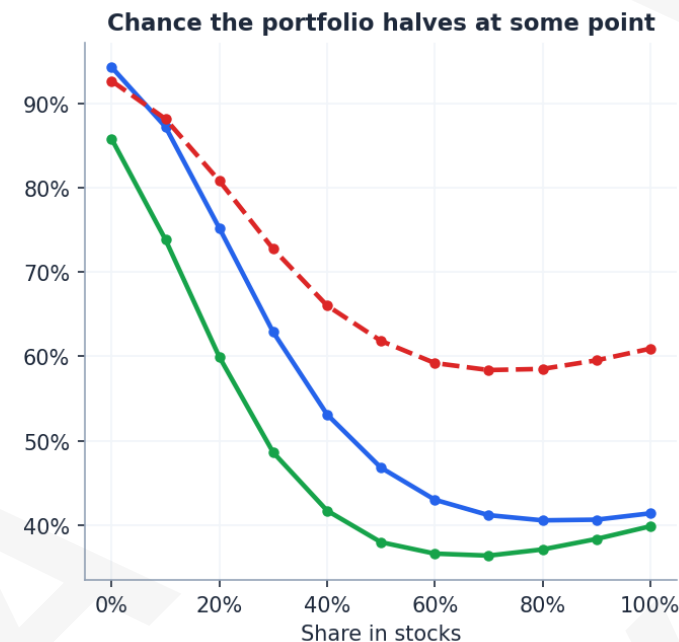
No stocks is the worst mix

Ruin of 62%. Bonds earn about 1.9% after inflation, short of a 4% draw.



50% to 80% stocks is flat

Ruin between 24.8% and 25.7%. All stocks lifts it to 27.0%.



A bear start moves the best mix

Lowest ruin at 50% stocks (39.5%). At 100% stocks it is 45.9%.

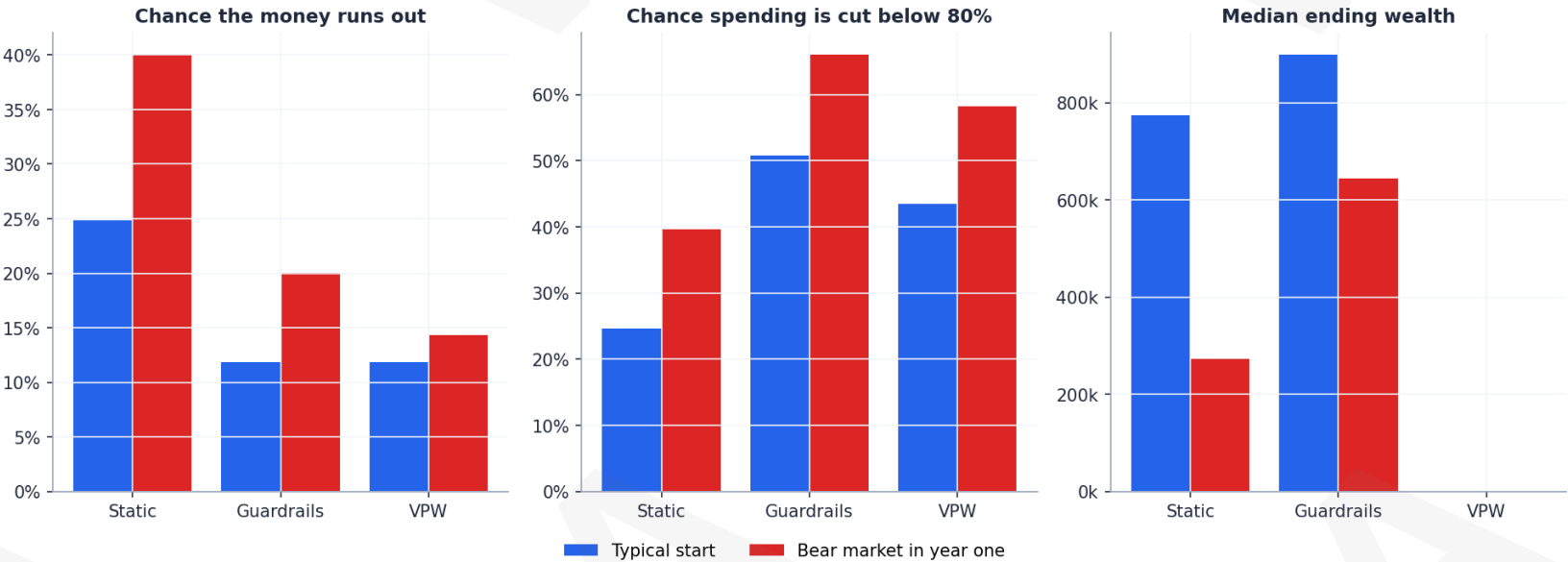
The best mix depends on how stretched the plan is

Assumptions	Best static mix	Static ruin there	Best guardrail mix	Guardrail ruin there
Default	70%	24.8%	40%	10.7%
Lower returns	80%	36.5%	50%	20.0%
Normal returns model	70%	21.3%	40%	6.4%
Spending \$50,000	100%	38.8%	70%	24.4%
Retire at 60	80%	30.0%	50%	16.2%

A plan that draws more than a safe portfolio earns cannot afford the slow drain of bonds. It has to take equity risk. Guardrails pull the best mix back to between 40% and 70% stocks, and beat static spending at every weight in every case.

Flexible spending is worth more than any weight change

Dynamic withdrawal rules move the risk. They do not remove it.



VPW spends the pot down by design, so its ending wealth is close to zero. Judge it on spending, not on what is left.

25% to 12%

Ruin with guardrails against static spending.

51% see a cut

Paths where spending drops below 80% of target at some point.

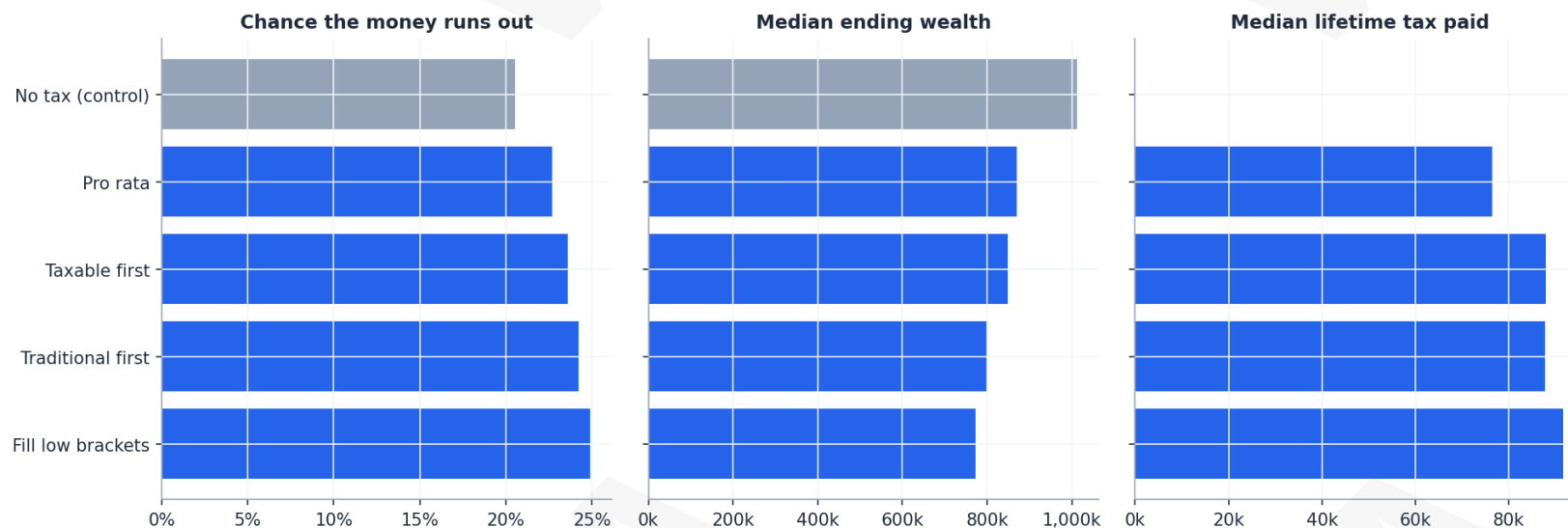
VPW spends the pot

Judge it on spending. Late care bills cause its ruin cases.

Static spending only falls short once the money has gone, so its cut rate (25%) is a ruin rate in disguise.

Tax order is a second order choice

Withdrawal order changes ruin risk by a few points, not by a lot



Tax adds 2 to 4 points

Against a no tax control.

Pro rata did best. 22.7%

Lowest lifetime tax, about \$76,000.

Low brackets did worst. 24.9%

Early tax gives up years of deferred growth.

What it implies, and what could change it

What it implies

Allocation has a wide, flat middle. Avoid the extremes.

All bond portfolios fail.

A stretched plan needs more stocks, not fewer.

A flexible spending rule is worth more than any weight change.

The first five years are the dangerous window.

What could change it

Lower returns lift base case ruin from 24.9% to 37.9%.

The regime numbers are judgement, not fitted estimates.

The plan runs to a fixed age. Nobody dies early.

One care episode at most. Same mix in every account.

No Social Security, pension or state tax.